

Math 104 – Real Analysis

Chapter-by-chapter notes on the important ideas

Ross, *Elementary Analysis* (2nd ed.) ■ Fall 2026 ■ Keshav Ramamurthy

Taking notes

Find the chapter banner, then type under the section heading as you read:

`\notesec{1A}{Span and Linear Independence}`

Record big ideas with `idea`, definitions with `definition`, theorems with `theorem`, `lemma`, `proposition`, examples with `example`, and warnings with `remark` — all numbered within the chapter. Add a missing section by copying a `\notesec` slot; add a chapter with `\chapterbanner`.

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Part I Introduction

Ross: the natural numbers, the rationals and their gaps, the reals, the completeness axiom

§1 The Set $\mathbb{N}$ of Natural Numbers**§2 The Set $\mathbb{Q}$ of Rational Numbers****§3 The Set $\mathbb{R}$ of Real Numbers****§4 The Completeness Axiom****§5 The Symbols $+\infty$ and $-\infty$** **§6 * A Development of $\mathbb{R}$**

Part II Sequences

Ross: limits, limit theorems, monotone and Cauchy sequences, subsequences, $\limsup$ and $\liminf$, series, convergence tests

§7 Limits of Sequences**§8 A Discussion about Proofs****§9 Limit Theorems for Sequences****§10 Monotone Sequences and Cauchy Sequences****§11 Subsequences****§12 $\limsup$'s and $\liminf$'s****§13 * Some Topological Concepts in Metric Spaces****§14 Series****§15 Alternating Series and Integral Tests****§16 * Decimal Expansions of Real Numbers**

Part III Continuity

Ross: continuous functions, the IVT and extreme values, uniform continuity, limits of functions

§17 Continuous Functions**§18 Properties of Continuous Functions****§19 Uniform Continuity****§20 Limits of Functions****§21 * More on Metric Spaces: Continuity****§22 * More on Metric Spaces: Connectedness**

Part IV Sequences and Series of Functions

Ross: power series, uniform convergence, the Weierstrass M -test, term-by-term calculus, approximation

§23 Power Series**§24 Uniform Convergence****§25 More on Uniform Convergence****§26 Differentiation and Integration of Power Series****§27 * Weierstrass's Approximation Theorem**

Part V Differentiation

Ross: derivative rules, the chain rule, the Mean Value Theorem, L'Hospital's rule, Taylor's theorem

§28 Basic Properties of the Derivative

§29 The Mean Value Theorem

§30 * L'Hospital's Rule

§31 Taylor's Theorem

Part VI Integration

Ross: the Riemann integral, its properties, the Fundamental Theorem of Calculus

§32 The Riemann Integral

§33 Properties of the Riemann Integral

§34 Fundamental Theorem of Calculus

§35 * Riemann–Stieltjes Integrals

§36 * Improper Integrals

Part VII Capstone

Ross: a discussion of exponents and logarithms, continuous nowhere-differentiable functions

§37 * A Discussion of Exponents and Logarithms

§38 * Continuous Nowhere-Differentiable Functions
